# Radioactive Decay and Half-Life

Answer key with worked solutions

## A. Understanding

**1.** Definitions.

> (a) The spontaneous decay of **unstable nuclei**, which emit radiation (alpha, beta, positron or gamma). (b) The **time for half** of the radioactive nuclei in a sample (and its activity) to decay. (c) The decay of any one nucleus is **random**, but in a very large number of nuclei a constant **fraction** decays in each unit of time, so the sample as a whole follows the half-life.

## B. Nuclei and nuclear equations

**2.** Protons and neutrons.

> (a) **92 p, 143 n** (b) **27 p, 33 n** (c) **53 p, 78 n** (d) **86 p, 136 n**. Neutrons = A − Z.

**3.** Completed equations.

> (a) $^{222}_{86}\mathrm{Rn}$ (b) $^{131}_{54}\mathrm{Xe}$ (c) $^{18}_{8}\mathrm{O}$ (d) $^{60}_{28}\mathrm{Ni}$ (e) $^{235}_{92}\mathrm{U}$ (f) $^{40}_{20}\mathrm{Ca}$. In each, the mass numbers and atomic numbers balance; for example (e): 239 = 235 + 4 and 94 = 92 + 2.

**4.** Identify X.

> (a) A = 0, Z = −1: a **beta particle**, $^{0}_{-1}\mathrm{e}$. (b) A = 4, Z = 2: an **alpha particle**, $^{4}_{2}\mathrm{He}$. (c) A = 0, Z = +1: a **positron**, $^{0}_{+1}\mathrm{e}$.

**5.** Uranium-238 series.

> $^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$; $^{234}_{90}\mathrm{Th} \longrightarrow {}^{234}_{91}\mathrm{Pa} + {}^{0}_{-1}\mathrm{e}$; $^{234}_{91}\mathrm{Pa} \longrightarrow {}^{234}_{92}\mathrm{U} + {}^{0}_{-1}\mathrm{e}$. Final nucleus: **uranium-234**, $^{234}_{92}\mathrm{U}$ (an isotope of the starting element).

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

## C. Types of radiation

**6.** Comparing radiation.

> **Alpha:** a helium nucleus, charge +2; stopped by paper or skin. **Beta:** an electron, charge −1; stopped by a few mm of aluminium. **Gamma:** electromagnetic radiation, no charge; only reduced by thick lead or concrete. Inside the body, an alpha emitter deposits all its energy in a very short distance in living cells, and alpha particles ionize strongly, so they cause the most damage there.

## D. Half-life

**7.** Technetium-99m.

> $n = \dfrac{24.04\ \text{h}}{6.01\ \text{h}} = 4.00$ half-lives
> activity $= 800\ \text{MBq} \times \left(\tfrac{1}{2}\right)^{4.00} = \dfrac{800\ \text{MBq}}{16} = \mathbf{50.0\ MBq}$

**8.** Half-life from data, and radon.

> (a) 1600 Bq → 800 Bq → 400 Bq is **2 half-lives**, so $t_{1/2} = \dfrac{30.0\ \text{min}}{2} = \mathbf{15.0\ min}$
> (b) $n = \dfrac{10.0\ \text{d}}{3.82\ \text{d}} = 2.618$; fraction remaining $= \left(\tfrac{1}{2}\right)^{2.618} = 0.163$, so **16.3 %** remains.

## E. Challenge

**9.** Radiocarbon age.

> $\dfrac{N_0}{N} = \dfrac{0.230\ \text{Bq/g}}{0.0850\ \text{Bq/g}} = 2.706$ (the units cancel)
> $n = \dfrac{\ln 2.706}{\ln 2} = \dfrac{0.9955}{0.6931} = 1.436$ half-lives
> $t = 1.436 \times 5730\ \text{yr} = \mathbf{8.23 \times 10^{3}\ yr}$ (about 8230 years). Check: the activity has fallen to 37.0 %, between 50 % and 25 %, so the age lies between 1 and 2 half-lives. ✓

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.